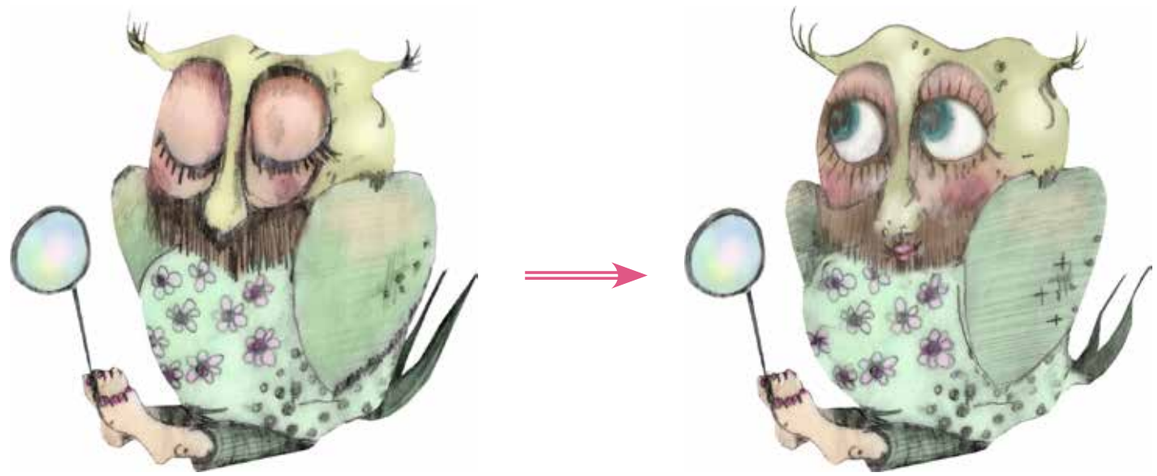
Image: hall_day.png

Bakgrund: Hall

Bakgrund: 2732 × 2048 px
Export: 72 PPI, PNG-24
OBS: SpriteKit Y = 2048 - PSD Y

Sequence 1: Sleeping owl



Övergång: Sleeping → Awake

SEQUENCE 1: SLEEPING OWL (IDLE)FILES: owl_sleep_01.
png → owl_sleep_05.png
Canvas: 1200 × 1200 px (all frames identical size)
PSD: X=2060, Y=1111, W=1200, H=1200
Code: x=2060, y=937 (SpriteKit: 2048-1111)
Scale: 0.75 (adjusted in code for proper size)TIMING:

Duration: 4 sek total per cycle
Frame rate: 0.5 sek per frame
Loop: Ping-pong (01→05→04→03→02→01) = 8 frames
Easing: Linear (steady, calm breathing)
SMOOTH INTERPOLATION: ← NYA!

Type: Cross-fade between frames
Fade out: 0.1 sek to alpha 0.9
Fade in: 0.1 sek to alpha 1.0
Hold: 0.3 sek at full opacity (0.5 - 0.2)
Method: Sequence (fadeOut, setTexture, fadeIn, hold)
Result: Silky smooth breathing (not choppy!)
Timing mode: Linear
LJUD:

File: TBD_owl_breathing.mp3
Trigger: Loop start (synced with animation)
Volume: 0.3 (subtle, ambient)
Loop: Continuous (matches animation)
Fade: None (already looping smoothly)
Notes: Soft ambient breathing sound
STATES:

State: Sleeping (initial/reset state)
Previous: None (starting state) or returned from pop
Next: Awake (on Tap 1)

Reset: On app launch or door exit
HAPTIC:

None (idle state, peaceful)
INTERAKTION:

Tap 1: Wake transition
TAP AREA:

Size: 650 × 550 px (generous for children)
Centered on sprite
Debounce: 0.3 sek (prevent rapid tapping)
EDGE CASES:

Rapid tap: Debounce 0.3s (ignore extra taps)
Background: Pause animation if app backgrounds
Low power: Reduce frame rate to 0.7s (optional)
VISUAL EFFECTS:

Glow day: None
Glow night: Very subtle (0.1 opacity, optional)
Eyes: Closed (visible in all frames)
Wings: Contracted (breathing motion)
Shadow: Code-generated under sprite (dynamic)
PERFORMANCE:

Preload: All 5 frames cached on scene init
Memory: ~600 KB total (5 frames × ~120 KB each)
FPS target: 60 fps (smooth cross-fade)
AFTER THIS SEQUENCE:

Remains: Sleeping idle loop continues
Next interaction: Tap 1 (wake up)
Continues: Until user taps or scene exits

TRANSITION: SLEEPING → AWAKE (WAKE UP)
Trigger: Tap 1 on sleeping owl
Duration: 1.5 sek total
Frame rate: 0.15 sek per frame
Frames: 10 frames total
FRAME SEQUENCE:
sleep_05 → sleep_04 → sleep_03 → sleep_02 → sleep_01
↓
awake_05 → awake_04 → awake_03 → awake_02 → awake_01

(5 sleep frames reversing + 5 awake frames reversing = 10 total)
SMOOTH INTERPOLATION: ← NYA!

Type: Cross-fade between frames
Fade out: 0.05 sek to alpha 0.88
Fade in: 0.05 sek to alpha 1.0
Hold: 0.05 sek at full opacity (0.15 - 0.1)
Method: Faster cross-fade (quicker transition)
Result: Smooth but energetic wake-up
Timing mode: EaseOut (accelerates out of sleep)

MOVEMENT:

Eyes: Gradually open (visible in frame progression)
Wings: Gradually expand outward
Body: Shifts from relaxed → alert posture

LJUD:

File: TBD_owl_wake_whoosh.mp3
Trigger: Frame sleep_05 (transition start)
Volume: 0.6
Loop: Nej (one-shot)
Fade: None (quick sound effect)
Duration: ~1.0 sek (shorter than animation)
Notes: Gentle "whoosh" or "flutter" sound

HAPTIC:

Type: Light impact
Trigger: Transition start (with tap)
Notes: Subtle feedback when user taps

STATES:

From: Sleeping (idle breathing)
To: Awake (alert idle)
Trigger: Tap 1 on owl tap area
Return: None (stays awake until bubble sequence completes)

VISUAL EFFECTS:

Glow: None in day mode
Glow: Subtle increase in night mode (0.1 → 0.3 opacity)

EDGE CASES:

Rapid tapping: Debounce 0.3 sek (ignore taps during transition)
Interrupt: If scene exits, complete current frame then stop
Background: Pause if app backgrounds during transition

PERFORMANCE:

Preload: All 10 frames cached before transition
Memory: ~1.2 MB (10 frames × ~120 KB each)
FPS target: 60 fps (smooth cross-fade timing)

AFTER TRANSITION:

Auto-start: Awake idle animation (sequence 2)
State: Set to "Awake"
Next interaction: Tap 2 (blow bubble)

↓ TECHNICAL NOTES - SMOOTH INTERPOLATION:
WHY CROSS-FADE MATTERS:
Without interpolation:
Frame 1 [SNAP] Frame 2 [SNAP] Frame 3
= Choppy, stop-motion feel

With cross-fade:
Frame 1 [smooth blend] Frame 2 [smooth blend] Frame 3
= Silky, professional animation
PARAMETERS USED:
Sleeping (slow, calm):

Fade duration: 0.1 sek (longer, gentler)
Fade depth: Alpha 0.9 (subtle fade)
Hold time: 0.3 sek (calm breathing)

Awake (faster, alert):

Fade duration: 0.08 sek (quicker)
Fade depth: Alpha 0.92 (more subtle)
Hold time: 0.14 sek (faster breathing)

Wake transition (energetic):

Fade duration: 0.05 sek (quick)
Fade depth: Alpha 0.88 (more visible fade)
Hold time: 0.05 sek (rapid sequence)

METHOD:
swift// Example smooth frame transition:
let fadeOut = SKAction.fadeAlpha(to: 0.9, duration: 0.1)
let setTexture = SKAction.setTexture(nextTexture)
let fadeIn = SKAction.fadeAlpha(to: 1.0, duration: 0.1)
let hold = SKAction.wait(forDuration: 0.3)
let smoothStep = SKAction.sequence([fadeOut, setTexture, fadeIn, hold])
RESULT:

Premium feel (AAA game quality)
Smooth, flowing animation
Not choppy or jarring
Professional polish

Rum: Hall
Objekt: Uggla

Sequence 2: Sleeping awake

Sequence 2: Bubbles

SEQUENCE 2: AWAKE OWL (IDLE)
FILES: owl_awake_01.png → owl_awake_05.png
Canvas: 1200 × 1200 px (all frames identical size)
PSD: X=2060, Y=1111, W=1200, H=1200
Code: x=2060, y=937
Scale: 0.75 (same as sleeping)
TIMING:

Duration: 3 sek total per cycle
Frame rate: 0.3 sek per frame
Loop: Ping-pong (01→05→04→03→02→01) = 8 frames
Easing: Ease-in-out (slightly more energy than sleeping)

SMOOTH INTERPOLATION: ← NYA!

Type: Cross-fade between frames
Fade out: 0.08 sek to alpha 0.92
Fade in: 0.08 sek to alpha 1.0
Hold: 0.14 sek at full opacity (0.3 - 0.16)
Method: Sequence (fadeOut, setTexture, fadeIn, hold)
Result: Smooth alert breathing (faster but still smooth)
Timing mode: EaseInEaseOut

LJUD:

File: TBD_owl_alert_ambient.mp3
Trigger: After wake transition completes
Volume: 0.4 (slightly more present than sleeping)
Loop: Continuous
Fade: Fade-in 0.5 sek (smooth transition from wake)
Notes: Subtle alert sounds, occasional soft hoot

STATES:

State: Awake (alert idle)
Previous: Sleeping (via wake transition)
Next: Blowing bubble (on Tap 2)
Return: After bubble pops (reset cycle)

HAPTIC:

None (idle state)

INTERAKTION:

Tap 2: Blow bubble

TAP AREA:

Size: 650 × 550 px (same as sleeping)
Centered on sprite
Debounce: 0.3 sek

EDGE CASES:

Rapid tap: Debounce 0.3s
Background: Pause animation

VISUAL EFFECTS:

Glow day: None
Glow night: Subtle increase (0.2 → 0.4 opacity)
Eyes: Fully open (visible in all frames)
Wings: Slightly more expanded than sleeping
Shadow: Same as sleeping (code-generated)

PERFORMANCE:

Preload: All 5 frames cached
Memory: ~600 KB total
FPS target: 60 fps

AFTER THIS SEQUENCE:

Remains: Awake idle loop continues
Next interaction: Tap 2 (blow bubble)
Or: Auto-sleep after timeout (optional, not implemented yet)

TRIGGER: Tap 2 på Awake Owl

OWL TREMBLING (under bubble grow):
Duration: 2.5 sek (simultant med bubble grow)
Movement: Subtle vibration/shiver
Horizontal: ±3-5 px (small jitter)
Vertical: ±2-3 px
Rotation: ±1-2° (barely noticeable)
Speed: 0.1 sek cycle (quick trembles)
Easing: Linear (constant shaking)
Amplitude: Increases slightly as bubble grows
Notes: Owl "strains" as blowing bigger bubble
VISUAL:

Eyes: Stay open (awake state maintained)
Beak: Can stay as-is (blowing position)
Body: Gentle continuous shake (effort of blowing)
Wings: Slight flutter (optional)

SOUND (optional):

File: TBD_owl_blowing_strain.mp3
Type: Gentle huffing/puffing (loop)
Volume: 0.3 (subtle background)
Duration: 2.5 sek (matches bubble grow)
Notes: Sounds like blowing hard

BUBBLE GROW ANIMATION:
FILES: bubble_01.png → bubble_08.png
Duration: 2.5 sek total
Frame rate: ~0.3 sek per frame
POSITIONS & SIZES:
bubble_01: 285×265px, x=1831, y=1207 → Code: x=1831, y=841
bubble_02: 298×331px, x=1720, y=1050 → Code: x=1720, y=998
bubble_03: 393×400px, x=1442, y=892 → Code: x=1442, y=1156
bubble_04: 530×492px, x=1206, y=676 → Code: x=1206, y=1372
bubble_05: 705×696px, x=963, y=418 → Code: x=963, y=1630
bubble_06: 968×920px, x=618, y=224 → Code: x=618, y=1824
bubble_07: 1749×1553px, x=333, y=22 → Code: x=333, y=2026
bubble_08: 2732×2048px, x=0, y=0 → Code: x=0, y=2048
INTERPOLATION:

Cross-fade: 0.1 sek fade, alpha 0.9
Position tween: Smooth curve between points
Easing: Ease-out (faster start, slower finish)

PATH:

Start: Near owl's beak (right side)
Movement: Dramatic left + upward curve
End: Fills entire screen (2732×2048)

SOUND:

File: TBD_owl_puff.mp3
Trigger: Tap 2 (bubble starts)
Volume: 0.7
Duration: ~1.0 sek

SIMULTANEOUS ACTIONS:

Bubble grows (01→08, 2.5s)
Bubble moves (along path positions)
Owl trembles (continuous shake)
All three happen together!

BUBBLE IDLE (wobble/anticipation):
Trigger: After bubble_08 reaches full size
Duration: Continuous loop (until Tap 3)
OWL STATE (during bubble wobble):

Trembling: STOPS (owl relaxes)
Position: Returns to stable awake idle
Animation: Gentle breathing (subtle)
State: Awake but calm (finished blowing)

Eyes: Stay open, watching bubble

BUBBLE WOBBLE:

Speed: 0.15 sek per cycle (fast vibration)
Horizontal: ±8-12 px (random variation)
Vertical: ±5-8 px
Rotation: ±2-4° (slight wiggle)
Scale pulse: 0.98 ↓ 1.02 (breathing)
Intensity: Optional build-up over 3-6 sek
Visual: Bubble looks "ready to burst"

SOUND (optional):

File: TBD_bubble_wobble_ambient.mp3
Type: Subtle squeaky/stretching (loop)
Volume: 0.2 (very quiet)
Stops: When tapped

TAP AREA:

Size: Entire screen (2732×2048)
Easy target: Anywhere triggers pop

SPECTACULAR FIREWORKS EXPLOSION (Tap 3):
Type: Grand finale (Chinese firework style)
Total duration: ~3.5 sek

Phase 1 - Pre-burst (0.2s):
- Wobble stops, 3 pulses, pause, tension build
- Sound: Fuse sizzle, Haptic: 3 light taps

Phase 2 - Initial boom (0.3s):
- White flash, 30-40 large particles (gold/white/silver)
- Size: 50-80px, velocity: 600-800 px/sec
- Sound: Deep boom (vol 1.0), Haptic: Heavy impact

Phase 3 - Cascading bursts (0.5s):
- Each particle splits into 3-5 smaller (120-160 total!)
- Colors: gold → rainbow spectrum
- Size: 20-40px, velocity: 300-500 px/sec
- Sound: Rapid crackling, Haptic: 10 medium impacts

Phase 4 - Sparkle trails (0.8s):
- Glowing trails (30-50px) behind particles
- 200-300 tiny sparkles added
- Twinkling effect (alpha pulse)
- Sound: Chimes/twinkles, Haptic: 20-30 light taps

Phase 5 - Gravity fall (1.5s):
- All particles + trails fall with physics
- Rotation, fade, trails persist longer
- Final sparkles last 0.5s

Visual: Screen shake, white flash, background dim
Result: SPECTACULAR grand finale effect!
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TOTAL DURATION: ~1.5 sek (explosion + fall)
OWL DURING EXPLOSION:

Position: Stable (no movement)
State: Awake, watching fireworks
Eyes: Can blink once (optional)
Reaction: Calm, satisfied expression

SOUND:

Pop: TBD_bubble_pop_big.mp3 (vol 0.9)
Sparkle: TBD_sparkle_twinkle.mp3 (vol 0.5)

HAPTIC:

Explosion: Heavy impact (maximum)
Sparkle: Light impact (optional)

CALM PAUSE:
Duration: 0.3 sek
Trigger: After particles fully faded (1.5s after pop)
STATE:

Hall: Clean (all particles gone)
Owl: Awake idle (calm)
Bubble: Removed from scene
Silence: Brief peaceful moment

OWL SLEEP TRANSITION:
Trigger: After 0.3s calm pause
Duration: 2.0 sek total
Type: Reverse of wake transition
FRAME SEQUENCE:
awake_01.png → awake_02 → awake_03 → awake_04 → awake_05
↓
sleep_05.png → sleep_04 → sleep_03 → sleep_02 → sleep_01

Total: 10 frames
Frame rate: 0.2 sek per frame
MOVEMENT:

Eyes: Gradually close (visible in frames)
Wings: Gradually contract (tuck in)
Body: Settles down (relaxed position)
Posture: From alert → relaxed

INTERPOLATION:

Cross-fade: 0.08 sek fade, alpha 0.92
Easing: Ease-in (gradual slowing)
Result: Smooth, peaceful transition

SOUND:

File: TBD_owl_yawn_sigh.mp3
Trigger: Transition starts
Volume: 0.5
Duration: ~1.5 sek
Character: Gentle yawn/sigh (content, sleepy)

HAPTIC:

None (peaceful, calm)

FINAL STATE - SLEEPING IDLE:
Trigger: After sleep transition completes
State: Sleeping (initial/reset state)
ANIMATION:

Sequence: sleep_01 → 05 → 04 → 03 → 02 (ping-pong)
Duration: 4 sek per cycle
Frame rate: 0.5 sek per frame
Loop: Continuous (until next Tap 1)
Easing: Linear (steady breathing)

OWL:

Eyes: Closed (sleeping)
Position: Stable, relaxed
Movement: Gentle breathing motion
Ready for: Tap 1 (wake again)

SCENE:

Bubble: Removed/cleaned up
Hall: Back to initial state
Memory: All bubble textures can be released
Ready: Full cycle can repeat

↓ COMPLETE TIMING BREAKDOWN:
0.0s: Tap 2 on awake owl
0.0s: Bubble starts growing (bubble_01)
0.0s: Owl starts trembling (simultaneous)
2.5s: Bubble reaches full size (bubble_08)
2.5s: Owl stops trembling
2.5s: Bubble wobble starts (continuous)
2.5s+: Owl calm, watching bubble wobble
[Waiting for Tap 3...]

Tap 3: (anytime after 2.5s)
0.0s: Pre-burst (0.1s)
0.1s: Explosion + particle burst (0.3s)
0.4s: Particles falling (0.8-1.2s)
1.5s: All particles faded
1.5s: Calm pause (0.3s)
1.8s: Owl sleep transition starts (2.0s)
3.8s: Sleeping idle begins
3.8s+: Continuous breathing loop

TOTAL CYCLE: ~6-7 sek (from Tap 2 to sleeping)

↓ STATE DIAGRAM:
Awake Idle
↓
[Tap 2]
↓
Blowing (owl trembles + bubble grows, 2.5s)
↓
Bubble Wobble (owl calm, bubble shakes)
↓
[Tap 3]
↓
Fireworks Explosion (1.5s)
↓
Calm Pause (0.3s)
↓
Sleep Transition (2.0s)
↓
Sleeping Idle
↓
[Tap 1 → back to Awake]

↓ SPECIAL DETAILS:
OWL TREMBLING EFFECT:

Synchronized: Matches bubble growth exactly
Subtle: Not distracting, just adds life
Purposeful: Shows effort of blowing
Stops: When bubble complete (owl relaxes)

BUBBLE WOBBLE:

Tension build: Creates anticipation
Visual cue: "Tap me!"
Satisfying: Build-up to big payoff

FIREWORKS:

Spectacular: Rainbow particles
Physics: Realistic fall
Rewarding: Big satisfying climax

SLEEP RETURN:

Peaceful: Gentle yawn
Complete: Full circle back to start
Ready: Can play again immediately

↓ NOTES FOR IMPLEMENTATION:
MEMORY MANAGEMENT:

Release bubble textures after pop
Particle system: Use SKEmitterNode or custom
Owl trembling: Simple SKAction sequence
All animations: Smooth 60 fps target

AUDIO LAYERING:

Owl puff (starts)
Owl strain (during blow, optional)
Bubble wobble (ambient, quiet)
Pop explosion (loud)
Sparkle twinkle (magical)
Owl yawn (peaceful)

HAPTIC SEQUENCE:

Medium impact: Tap 2 (bubble starts)
Heavy impact: Tap 3 (explosion)
None: Sleep transition (peaceful)
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